PORTFOLIO

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INTRODUCTION

안녕하세요, 우영범입니다.

저는 Text-to-Image Diffusion 연구를 시작으로 멀티모달 AI와 AI Agent 시스템 개발까지 경험을 확장해왔습니다. 주요 관심 분야는 생성형 AI, Vision-Language Model, 이미지 생성 및 평가이며, 기존 기술의 한계를 분석하고 이를 알고리즘과 시스템 구조 수준에서 개선하는 데 집중하고 있습니다.

최근에는 생성 모델을 활용한 합성데이터 생성과 품질 평가 방법에도 관심을 넓히고 있습니다. 현재는 다중 개념 이미지 생성 연구에서 축적한 경험을 바탕으로, 이미지 이해부터 프롬프트 구조화, 생성, 평가, 개선까지 연결되는 멀티모달 AI 시스템을 개발하고 있습니다. 새로운 기술을 단순히 적용하는 데 그치지 않고, 실제로 작동하고 검증 가능한 형태로 구현하는 AI 엔지니어를 지향합니다.

EDUCATION

학력 사항

Korea University (PRML Lab)

M.S. in Artificial Intelligence, 2023.09 - 2026.02

Research Topics : Generative AI, Text-to-Image Diffusion, Multi-Concept Personalization

GPA: 3.94/4.5

Ajou University

B.S. in Cyber Security, 2016.03 - 2023.08

Main Courses : Artificial Intelligence, Machine Learning, Probability and Statistics, Big Data Security

GPA: 3.11/4.5

PUBLICATIONS

연구 활동

2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC 2025)

Young-Beom Woo, Sun-Eung Kim, Seong-Whan Lee,
"FlipConcept: Tuning-Free Multi-Concept Personalization for Text-to-Image Generation",
(Oral Presentation), (Oct. 2025)

IEEE SMC 2025
Systems, Man, and Cybernetics Society
Vienna, Austria

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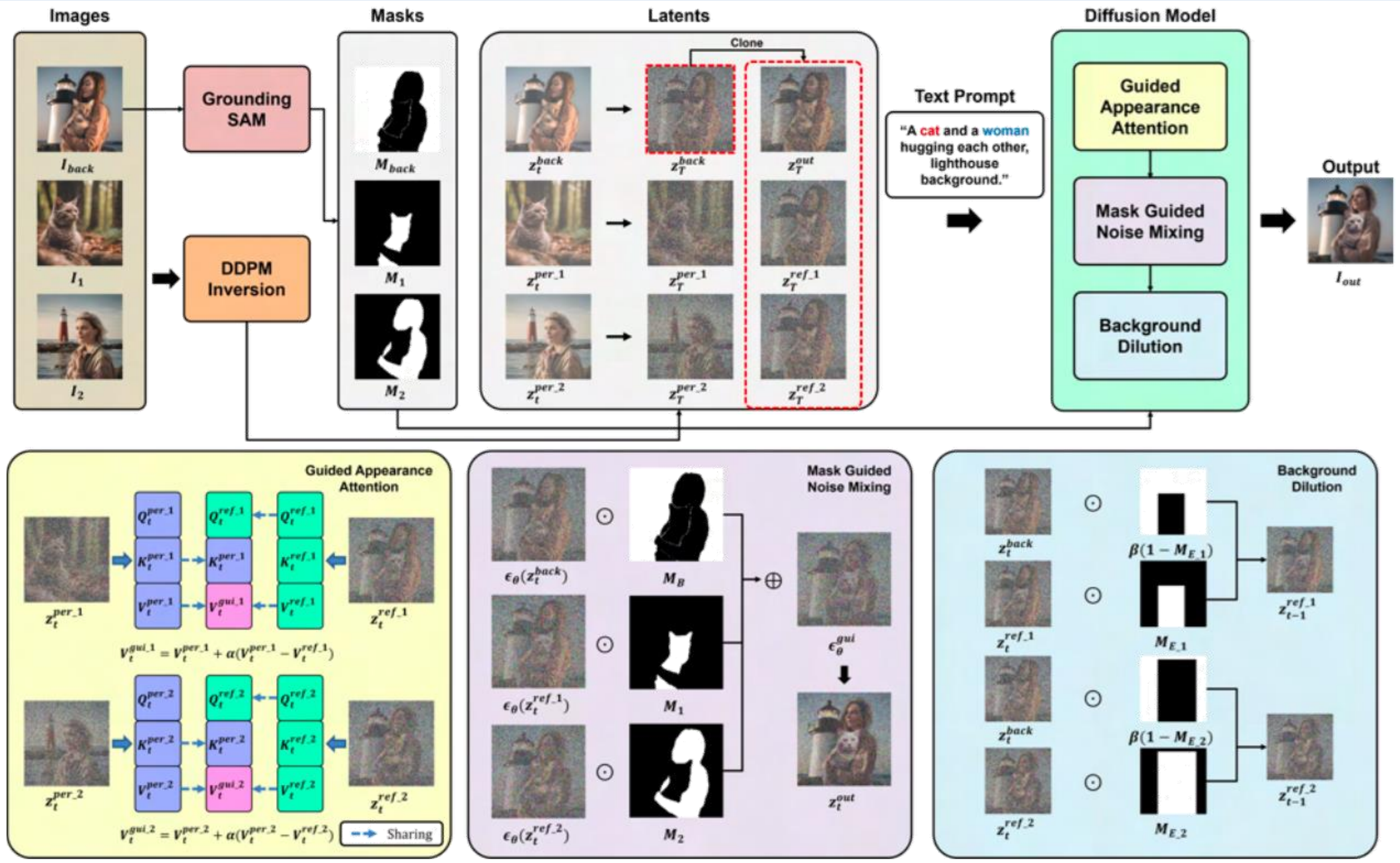


Overview of FlipConcept

We introduce FlipConcept, a tuning-free framework for multi-concept personalization that

1) *preserves the visual identity of each personalized concept through Guided Appearance Attention,*
2) *selectively blends object and background regions using Mask-Guided Noise Mixing, and*
3) *reduces concept leakage with Background Dilution.*

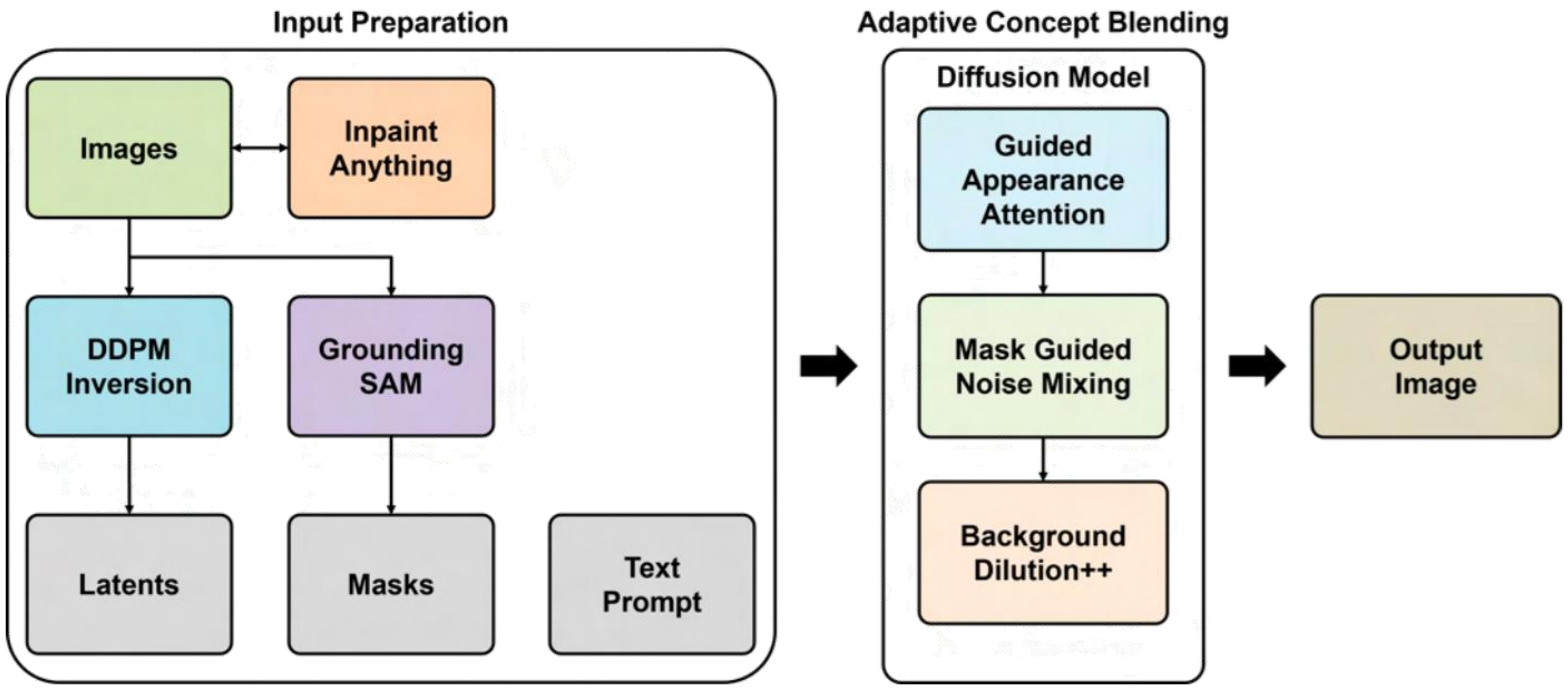
Without additional fine-tuning, FlipConcept generates coherent multi-object images while maintaining concept fidelity, spatial consistency, and background structure, achieving improved CLIP and DINO scores as well as substantially faster generation than tuning-based methods.

PUBLICATIONS

연구 활동

Master's Thesis, Korea University, 2026

Young-Beom Woo, Sun-Eung Kim, Seong-Whan Lee,
"Plug-and-Play Multi-Concept Adaptive Blending for High-Fidelity Text-to-Image Synthesis", (2026)



Overview of PnP-MIX

We introduce PnP-MIX, a plug-and-play framework for high-fidelity multi-concept personalization without additional model tuning. PnP-MIX first prepares editable latent representations and object masks from a user-selected scene, then adaptively blends multiple personalized concepts through Guided Appearance Attention, Mask-Guided Noise Mixing, and Background Dilution++. By preserving non-personalized regions and reducing the unintended leakage of visual attributes across objects and background regions, PnP-MIX produces coherent multi-object images with improved concept fidelity, boundary alignment, and background consistency.

PROJECTS

(2026)

Multimodal AI Agent Playground

: Reference-aware Multimodal AI Agent Framework with Structured Requirement Parsing and Multi-Metric Evaluation

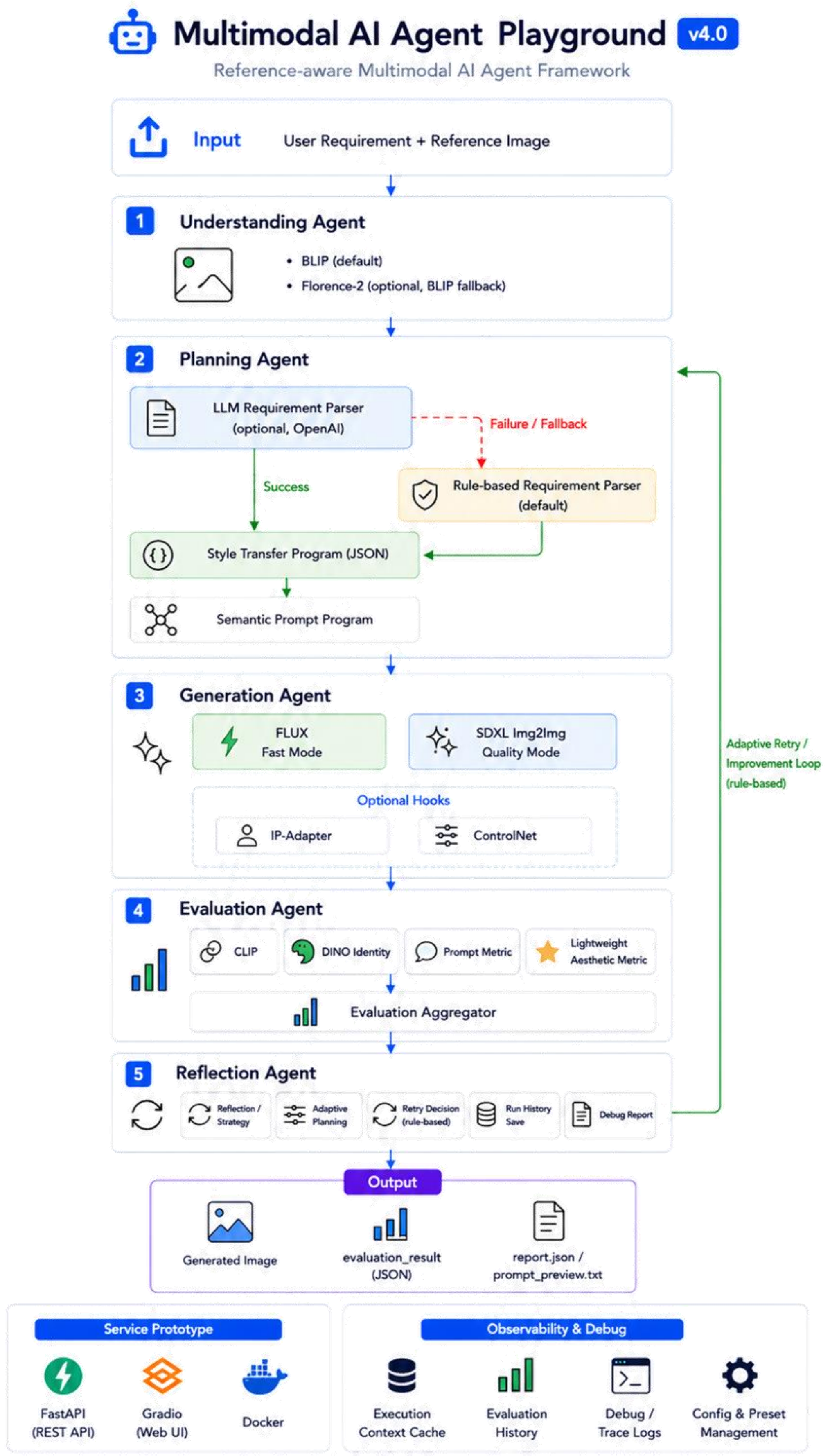
Tech Stack

- ▶ Vision & Generation: BLIP, FLUX, SDXL Img2Img
- ▶ LLM Integration: OpenAI API, LLM Requirement Parsing, Rule-based Fallback
- ▶ Evaluation: CLIP, DINO
- ▶ Framework & Service: PyTorch, Transformers, Diffusers, FastAPI, Gradio, Docker
- ▶ Development Workflow: OpenAI Codex, Git, GitHub
- ▶ Optional Integrations: Florence-2, IP-Adapter, ControlNet

My Role

- Multimodal AI Agent Framework Designer & Developer (*Individual Project*)
    - Designed a reference-aware multimodal AI framework connecting image understanding, structured planning, generation, evaluation, and iterative improvement.
    - Built a provider-independent vision pipeline and structured user requirements into Style Transfer and Semantic Prompt Programs.
    - Added optional LLM-based requirement parsing with rule-based fallback for stable execution.
    - Implemented provider-specific prompt rendering and FLUX/SDXL Img2Img generation routes.
    - Developed CLIP, DINO, prompt consistency, and lightweight aesthetic evaluation with rule-based retry logic.
    - Used Codex for iterative development and packaged the workflow as a FastAPI, Gradio, and Docker-based prototype with traceable debug reports.

Contribution ★★★★★



Overall Structure of Multimodal AI Agent Playground

Overall Architecture of Multimodal AI Agent Playground

PROJECTS

(2023)

A.C.G.C.

: GUI-based AI Code Generation Tool for Machine Learning

Tech Stack

▶ LLM Integration: OpenAI GPT API, Prompt Template Engineering

▶ ML Templates: LightGBM, XGBoost, Random Forest, Extra Trees

▶ Data Processing: Pandas, Scikit-learn, Optuna

▶ Application: Python, PyQt5

My Role (Team of 3)

- Prompt & ML Code Generation Developer
    - Designed prompt templates and code-generation logic for classification and regression tasks.
    - Converted dataset path, task type, and target column into structured GPT prompts using predefined ML templates.
    - Supported Python code generation for LightGBM, XGBoost, Random Forest, and Extra Trees models.
    - Integrated GPT calls with a PyQt5 GUI and refined prompts and exception handling through repeated script testing.

Contribution ★★★★☆

PROJECTS

(2022)

SPAM MAIL DETECTION

: Comparative Analysis of ML and DL Models for Spam Email Classification

Tech Stack

▶ Models: Logistic Regression, Random Forest, SVM, MLP, CNN

▶ ML Libraries: Pandas, Scikit-learn, Optuna

▶ Evaluation: Cross-Validation, F1-Score, ROC Curve, PR Curve

▶ Development: Python, PyTorch, Google Colab

My Role (Team of 3)

- ML Experiment Lead & Team Leader
    - Led project planning, experiment design, task allocation, and result integration.
    - Performed EDA, text preprocessing, and class imbalance handling on approximately 5,000 email samples.
    - Implemented and compared Logistic Regression, Random Forest, SVM, MLP, and CNN models.
    - Evaluated models using cross-validation, F1-score, ROC and PR curves, and analyzed their strengths and limitations.

Contribution ★★★★☆

Thank You!

우영범

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